WORKED SOLUTION

STEP III 1997, Question 4

Pure • AM-GM Inequality

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q4

The key idea

The question as printed has a misprint: a polynomial with four roots must be a quartic, and the pattern of coefficients $1, 4, 6, 4, 1$ tells us the intended polynomial is

$$\mathrm{f}(x) = x^4 - 4ax^3 + 6b^2x^2 - 4c^3x + d^4.$$

The coefficients are chosen so that each of a, b^2, c^3, d^4 is an *average*: an average of the roots, of the products of pairs, of the products of triples, and of the product of all four. Each inequality then comes from the AM–GM inequality applied to the roots of f, f′ or f″, with Rolle's theorem making sure those roots are still distinct and positive.

Solution

Write $\mathrm{f}(x) = (x-p)(x-q)(x-r)(x-s)$. Comparing coefficients,

$$p+q+r+s = 4a, \qquad pq+pr+ps+qr+qs+rs = 6b^2,$$
$$pqr+qrs+rsp+spq = 4c^3, \qquad pqrs = d^4.$$

Part (i)

Let $P = pqrs = d^4$. Since none of the roots is zero,

$$pqr = \frac{P}{s}, \quad qrs = \frac{P}{p}, \quad rsp = \frac{P}{q}, \quad spq = \frac{P}{r}.$$

The map $t \mapsto P/t$ is one-to-one on the positive reals, and p, q, r, s are distinct, so the four products are distinct.

Part (ii)

The four triple products are distinct positive numbers, so by the AM–GM inequality

$$c^3 = \frac{pqr+qrs+rsp+spq}{4} > (pqr \cdot qrs \cdot rsp \cdot spq)^{1/4} = ((pqrs)^3)^{1/4} = (d^4)^{3/4} = d^3.$$

Each root appears in exactly three of the four products, which is why the product is $(pqrs)^3$. Since $c, d > 0$ and $t \mapsto t^3$ is increasing, $c^3 > d^3$ gives $c > d$.

Part (iii)

Label the roots so that $0 < p < q < r < s$. The polynomial f is differentiable and vanishes at p and q, so by Rolle's theorem f′ has a root α with $p < \alpha < q$. In the same way there are roots $\beta \in (q, r)$ and $\gamma \in (r, s)$. These open intervals don't overlap, so $\alpha < \beta < \gamma$ are distinct. As f′ is a cubic it has no other roots. Note also that all three are positive, because $\alpha > p > 0$.

Part (iv)

Differentiating,

$$\mathrm{f}'(x) = 4x^3 - 12ax^2 + 12b^2x - 4c^3 = 4(x^3 - 3ax^2 + 3b^2x - c^3),$$

so α, β, γ are the roots of $x^3 - 3ax^2 + 3b^2x - c^3$, and

$$\alpha + \beta + \gamma = 3a, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = 3b^2, \qquad \alpha\beta\gamma = c^3.$$

This is the same pattern one degree down, so we copy parts (i) and (ii). The pair products are $\alpha\beta = c^3/\gamma$, $\beta\gamma = c^3/\alpha$, $\gamma\alpha = c^3/\beta$, which are distinct because α, β, γ are distinct and positive. By the AM–GM inequality,

$$b^2 = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{3} > ((\alpha\beta\gamma)^2)^{1/3} = c^2,$$

and since $b, c > 0$, we get $b > c$.

Ask yourself Why do we need part (iii) before we can use AM–GM on f′?

The AM–GM inequality only applies to positive reals, and we need the strict version. A cubic needn't have three real roots at all. Rolle's theorem is what guarantees that α, β, γ are real, positive and distinct, and so the pair products are too.

Part (v)

Apply the same argument once more. Since f′ has distinct roots $\alpha < \beta < \gamma$, Rolle's theorem gives roots $\mu \in (\alpha, \beta)$ and $\nu \in (\beta, \gamma)$ of

$$\mathrm{f}''(x) = 12x^2 - 24ax + 12b^2 = 12(x^2 - 2ax + b^2),$$

and these are distinct and positive. So $\mu + \nu = 2a$ and $\mu\nu = b^2$, and by the AM–GM inequality for two numbers,

$$a = \frac{\mu + \nu}{2} > \sqrt{\mu\nu} = b.$$

(Equivalently, $x^2 - 2ax + b^2$ has two distinct real roots, so its discriminant $4a^2 - 4b^2$ is positive, and $a > b$ because both are positive.) Altogether we have shown

$$a > b > c > d.$$

Checks

- Roots $1, 2, 3, 4$: then $4a = 10$, $6b^2 = 35$, $4c^3 = 50$, $d^4 = 24$, so $a = 2.5$, $b \approx 2.415$, $c \approx 2.321$, $d \approx 2.213$, in the right order.
- If all four roots were equal to k, then $\mathrm{f}(x) = (x-k)^4$ and $a = b = c = d = k$. That is why distinctness is needed for the strict inequalities.
- We tested 20 000 random sets of four distinct positive roots, including nearly coincident ones, and $a > b > c > d$ held every time.

Common slips

- Applying AM–GM to the roots of f directly. That gives $a > d$, but it can't separate the middle terms.
- Forgetting to check that the numbers fed into AM–GM are positive and distinct, which is the whole point of parts (i) and (iii).
- Going from $c^3 > d^3$ or $b^2 > c^2$ to $c > d$ or $b > c$ without noting that all the letters are positive.

Beyond STEP

The quantities a, b, c, d are the *symmetric means* of the roots, and the chain $a > b > c > d$ is a special case of *Maclaurin's inequality*: for positive reals, the k-th root of the average of the k-fold products decreases as k increases. The trick of differentiating to lower the degree while keeping the means unchanged is also the standard route to Newton's inequalities. For much more, see Steele, *The Cauchy–Schwarz Master Class*, or the classic by Hardy, Littlewood and Pólya, *Inequalities*.

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